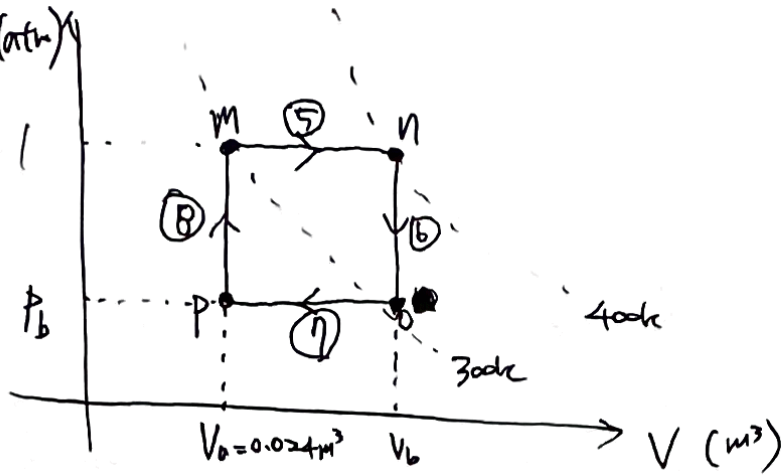


(2) P(atm)

(15)



⑤: $\frac{1}{2}$ of $W = \int_m^n P dV = -W_{\oplus} = 800 \text{ J}$

$\frac{1}{2}$ of $W = -800 \text{ J}$

③

⑥/⑧: $\Delta V = 0 \rightarrow \frac{1}{2}$ of $W = 0 \rightarrow \frac{1}{2}$ of $W = 0 \text{ J}$

⑦: $\frac{1}{2}$ of $W = P_b \times (V_p - V_0) = P_b (V_a - V_b)$

$V_a = 0.024 \text{ m}^3, V_b = ?$

$$V_b = \frac{nRT_1}{P_n} = \frac{1 \text{ mol} \times 8 \text{ J/mol} \cdot \text{K} \times 400 \text{ K}}{1 \text{ atm}} = \frac{1 \text{ mol} \times 8 \text{ J/mol} \cdot \text{K} \times 400 \text{ K}}{10^5 \text{ Pa}} = 0.032 \text{ m}^3$$

$\therefore W = P_b \times (0.024 - 0.032)$

$= -0.008 \times \frac{P_b}{\text{Pa}} \text{ J}$

$\frac{1}{2}$ of $W = +0.008 P_b \text{ J}$

$\frac{1}{2}$ of $W = 0.008 P_b - 800 \text{ J} = -560 \text{ J}$

$P_b = 30000 \text{ Pa} = 0.3 \text{ atm}$

③

③